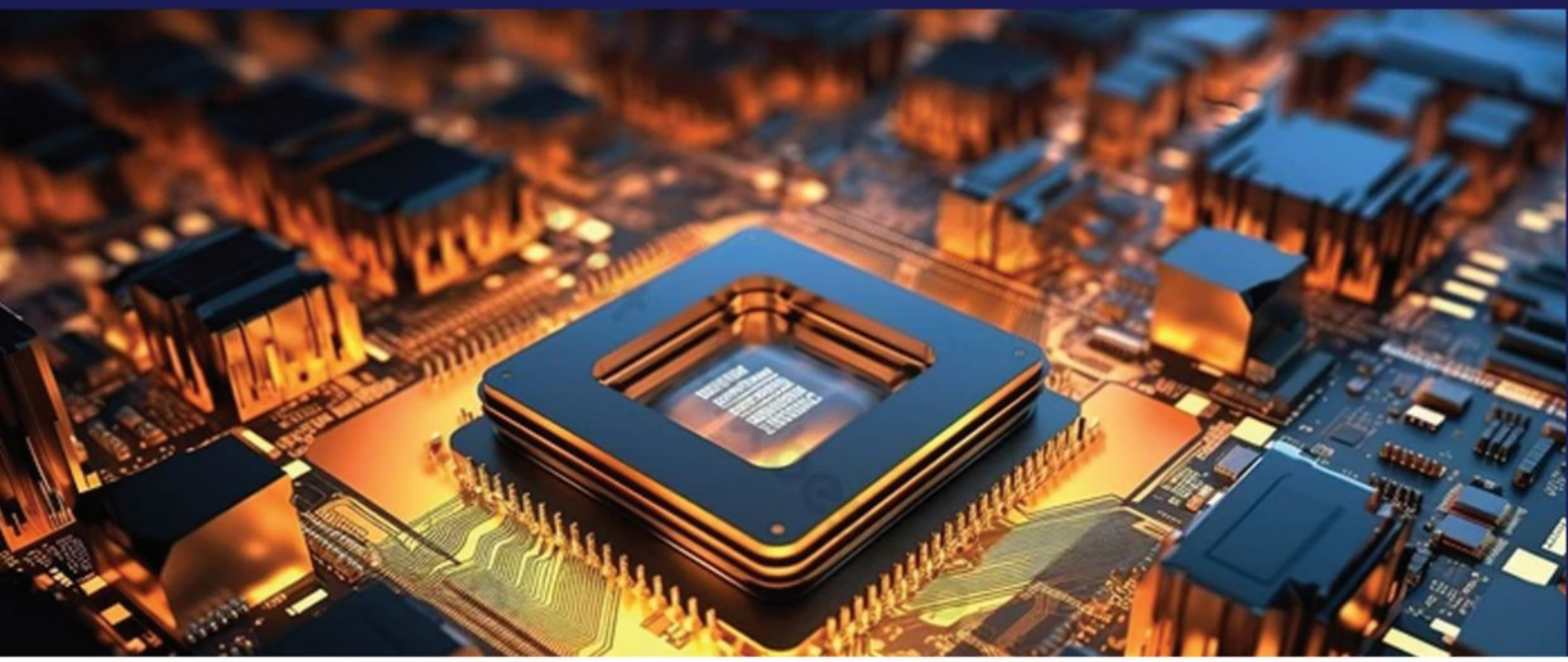
PROCEEDINGS OF THE INTERNATIONAL CONFERENCE ON

ADVANCED COMPUTING

ICAC 2024

26th and 27th September 2024



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Machine Learning and Artificial Intelligence in Predicting Disease Progression in ALS: A Comprehensive Review on Current Models and Future Directions

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Abstract---Amyotrophic Lateral Sclerosis (ALS) is a severe neurodegenerative disorder that progressively impairs motor neuron function, leading to muscle weakness, paralysis, and often, respiratory failure. Accurate prediction of ALS progression is vital for patient management and the design of effective clinical trials. The advent of machine learning (ML) and artificial intelligence (AI) has paved the way for sophisticated predictive models that leverage various data sources, including clinical records, neuroimaging, and biomarkers. This review comprehensively evaluates five pivotal studies that utilize algorithms such as Random Forest (RF), Support Vector Machines (SVM), Logistic Ridge Regression, and Deep Leaming to forecast ALS progression. We critically analyze the algorithms employed, the datasets used, and each model's performance in predicting survival times, motor function decline, and overall disease trajectory. Our findings indicate that integrating diverse features from clinical, biological, and neuroimaging data enhances predictive accuracy significantly. For instance, RF models achieved up to 83% accuracy, while SVM models reached up to 94% classification accuracy under specific conditions. We also outline prospective research avenues aimed at refining these predictive models and expanding their applicability to broader and more diverse patient populations.

Keywords---Amyotrophic Lateral Sclerosis (ALS), Machine Learning, Random Forest, Support Vector Machines, Deep Learning, Prognosis, Artificial Intelligence, Clinical Data, MRI, Biomarkers

I. INTRODUCTION

Amyotrophic Lateral Sclerosis (ALS) is a progressive neurodegenerative disorder characterized by the gradual loss of upper and lower motor neurons. This relentless decline leads to severe muscle weakness, progressive paralysis, and ultimately, respiratory failure, with a typical survival period ranging between 2 to 5 years post-diagnosis. Despite this general timeframe, the rate and pattern of disease progression can vary widely among individuals, making accurate predictions challenging. Recent strides in machine learning (ML) and artificial intelligence (AI) offer transformative potential to address this challenge[1]. These technologies enable the development of sophisticated predictive models that integrate various data types, such as clinical records, neuroimaging data, and biomarkers, to forecast ALS progression with greater accuracy. This paper aims to provide a thorough review of current methodologies employed in predicting ALS progression, focusing on different ML and AI algorithms such as Random Forest (RF), Support Vector Machines (SVM), Deep Leaming models, and Logistic Ridge Regression[2]. By examining how these algorithms utilize diverse

datasets-including clinical trial data from the PRO-ACT database, MRI imaging data, and blood biomarker profiles-we seek to identify the strengths and limitations of each approach and propose directions for future research to enhance predictive accuracy and applicability[6].

	Training	Test	Validation
Gender			
Female	53	182	107
Male	91	303	222
n.a.	135	433	296
Race			
Caucasian	0	8	5
Asian	4	14	9
Black/African Am.	272	886	598
n.a.	3	10	13
Onset site			
Bulbar	59	203	121
Limb	218	711	500
Limb and bulbar	2	4	4
Age (years)	53 (44–62)	55 (47–64)	56 (47–64)
Height (cm)	170 (164–178)	170 (163–178)	172 (165–179)
Weight (kg)	74 (63–86)	73 (60–85)	71 (60–84)
Number of visits	12 (11–12)	12 (11–12)	4 (3–4)
3–12 month period (days)	277 (258–305)	279 (260–303)	n.a.

Table 1. Dataset on Random Forest

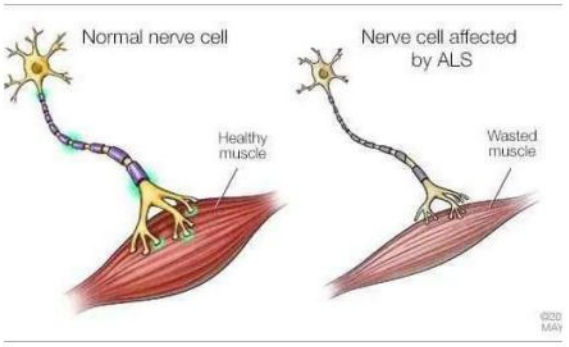


Fig1. ALS affected Nerve Cells

II. LITERATURE REVIEW

2.1 DATA SOURCES

2.1.1 RANDOM FOREST

The PRO-ACT (PROgressive multifocal leukoencephalopathy and Amyotrophic Lateral Sclerosis Clinical Trial) database is a crucial public resource that aggregates clinical trial data from over 10,000 ALS patients, offering a comprehensive array of variables including ALS Functional Rating Scale-Revised (ALSFRS-R) scores, Forced Vital Capacity (FVC) measurements, and detailed demographic information such as age, gender, and disease duration. This extensive dataset provides a robust foundation for developing predictive models, as it enables the analysis of various clinical factors and their impact on ALS progression[2]. By integrating longitudinal data on motor function, respiratory capacity, and other clinical measures, the PRO-ACT database supports the creation of sophisticated models for forecasting disease trajectories and enhances our understanding of ALS, ultimately contributing to more accurate and personalized patient management and research advancements[7].

2.1.2 MRI BASED DATASET

MRI datasets, including both structural and diffusion tensor imaging (DTI) data, were collected from ALS patients and healthy controls to investigate neuroanatomical changes associated with the disease. These datasets focus on critical brain regions affected by ALS, such as the precentral and paracentral gyri, which are involved in motor function; the corpus callosum, crucial for interhemispheric communication; and the corticospinal tracts, which are directly related to motor pathways[4]. By analyzing these neuroimaging features, researchers can identify distinct patterns of neurodegeneration and disruptions in connectivity associated with ALS. Such analyses are essential for understanding the disease's impact on brain structure and function, improving diagnostic accuracy, and developing targeted therapeutic interventions[3].

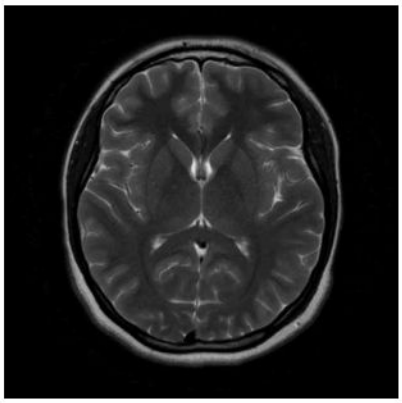


Fig2.Normal Brain on MRI scan

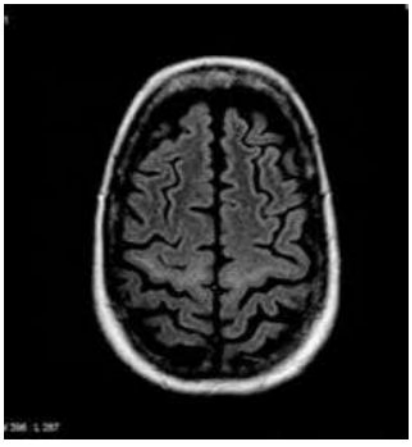


Fig3. Affected Brain on MRI scan

Table 2 Blood based biomarkers based on ALS categories (7)

Familial ALS Biomarkers	Sporadic ALS Biomarkers
TDP-43	TDP-43
FUS	FUS
C9ORF72	Neurofilaments
	Extracellular RNA
	Stress granules
	Progranulin
	TNF-a and range of cytokines
	Metabolites (i.e., creatine kinase, platelet serotonin)
	Mitochondrial biomarkers (i.e., cytochrome oxidase, mitochondrial respiration rate, reduced Complex-II activity)

2.1.3 BLOOD BIOMARKER DATA-SET

Blood biomarker data from ALS patients and individuals with Lower Motor Neuron Disease (LMND) were meticulously analyzed to identify a diverse range of biomarkers, including metabolic, oxidative, and inflammatory markers. These biomarkers offer valuable insights into the underlying biological processes and pathophysiological mechanisms of ALS, such as alterations in metabolic pathways, oxidative stress levels, and inflammatory responses[1]. By examining these biochemical profiles, researchers can better understand the disease's progression and its impact on systemic functions. This comprehensive biomarker data is instrumental in developing predictive models that leverage biochemical signatures to forecast disease trajectory, identify potential therapeutic targets, and enhance diagnostic accuracy, ultimately contributing to more effective management and treatment strategies for ALS[8].

Table 1 Blood Based Biomarker Studies

Target	Source	Sensitivity/Specificity	n
TDP-43	Plasma	NA	319
Exosome miRNA	Plasma	NA	40
Exosome proteomics	Plasma	NA	22
Proteomics	Plasma	58% and 90%	295
Glutamate Uptake	Platelet	NA	82
Mito-Respiration	Platelet	NA	15
Serotonin	Platelet	NA	114
NfL	Serum/Plasma	84–100% and 76–97%	248
NfH	Serum/Plasma	61–80%,72.1–83.7%	157–331
Cytokines	Serum/Plasma	NA	87–183
Ferritin	Serum/Plasma	NA	104–694
Creatine Kinase	Serum/Plasma	63.8% and 54.3%	216–834
Non-coding RNA	Whole Blood	73.9–93.7%	88
Chromosomal Confirmation	Whole Blood	83.33–87.5%	58
Microarray analysis	Whole Blood	87%	1,116
Immune Cell Profiling	Whole Blood	NA	80
T-regs	Whole Blood	73.9–76.9%, 69.6–73.1%	217

Table 2. Blood Biomarker Dataset

2.2 CURRENT MODELS

2.2.1 RANDOM FOREST (RF)

Random Forest (RF) is an ensemble learning method that builds multiple decision trees during training and aggregates their outputs to make predictions. This approach leverages the power of many decision trees to enhance prediction accuracy and robustness while reducing the risk of overfitting, making it highly effective for analyzing high-dimensional data. In the realm of ALS research, RF models have proven to be particularly valuable. When applied to the PRO-ACT dataset, RF models have achieved notable success in predicting survival times, with accuracies ranging from 79% to 83% depending on the combination of clinical features used[2].

2.2.3 DEEP LEARNING WITH NEURAL NETWORKS

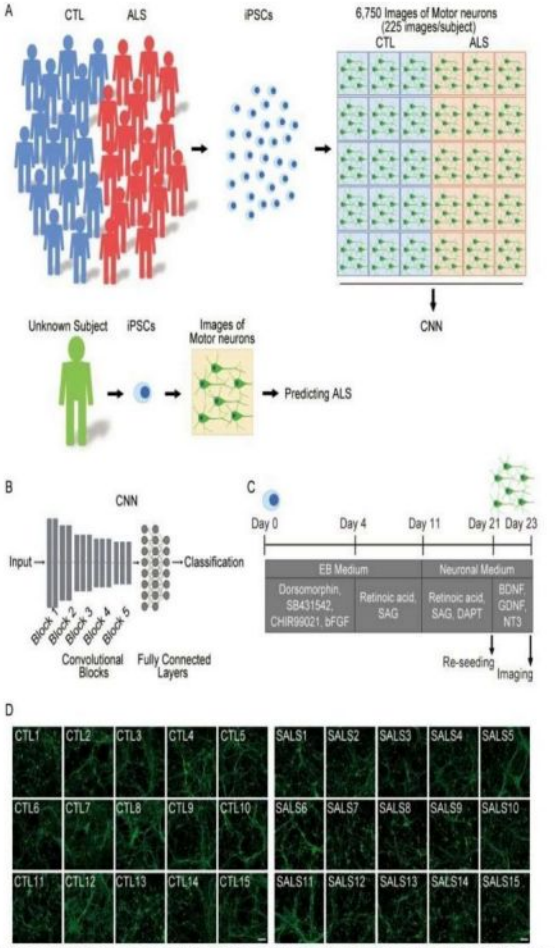


FIGURE 1: Amyotrophic lateral sclerosis (ALS) prediction model using induced pluripotent stem cell (iPSC)-derived motor neurons. (A) Schematic illustration of the prediction model for ALS using iPSC-derived motor neurons and convolutional neural network (CNN). (B) Schema of CNN. VGG-16 network pretrained by ImageNet dataset and substitute fully connected layers were recruited. (C) Differentiation schedule of motor neurons from iPSCs for imaging. (D) Representative images of motor neurons derived from iPSCs of healthy control (CTL) subjects and sporadic ALS (SALS) patients. Scale bars = 100μm. BDNF = brain-derived neurotrophic factor; bFGF = basic fibroblast growth factor; GDNF = glial cell line-derived neurotrophic factor; NT-3 = neurotrophin-3; SAG = smoothened agonist.

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This indicates the method's proficiency in handling and extracting meaningful patterns from complex clinical data. Furthermore, RF models applied to MRI data have demonstrated their versatility by achieving an accuracy of 79.17% in predicting 18-month survival, underscoring their ability to integrate and analyze diverse types of data such as neuroimaging features. This effectiveness highlights the potential of RF models to provide reliable predictions and support clinical decision-making in ALS, given their capacity to manage complex, multi-dimensional datasets and identify significant predictors of disease progression[7]

2.2.2 LOGISTIC RIDGE REGRESSION

Logistic Ridge Regression is a robust predictive technique that integrates clinical and neuroimaging data to forecast ALS survival outcomes. This method combines clinical variables, such as ALS Functional Rating Scale-Revised (ALSFRS-R) scores, with neuroimaging measures, such as cortical thickness obtained from MRI scans. By applying a ridge penalty to large regression coefficients, Logistic Ridge Regression mitigates the risk of overfitting, thereby enhancing the model's generalization capabilities and ensuring more reliable predictions. This approach achieved a prediction accuracy of 79.17% for 18-month survival, demonstrating its effectiveness in integrating diverse data sources to forecast disease progression[5]. Additionally, the model exhibited a specificity of 83.34% and a sensitivity of 75% for short-term survival predictions. These metrics underscore the model's practical utility, as it effectively balances the identification of true positives and true negatives, making it a valuable tool for predicting both short-term and longer-term outcomes in ALS. The combination of clinical and imaging data, coupled with the regularization provided by ridge regression, enhances the model's robustness and reliability, offering significant insights for patient management and treatment planning[2].

Deep Leaming models, especially Convolutional Neural Networks (CNNs), have shown significant promise in analyzing complex ALS datasets by uncovering intricate patterns in both clinical and imaging data. CNNs are particularly adept at feature extraction from large volumes of data, which enhances their ability to make accurate survival predictions. They leverage their multiple layers to automatically learn and represent hierarchical features, making them highly effective for tasks involving image and multidimensional data[3]. CNNs trained on MRI data have demonstrated impressive performance, achieving an accuracy of 84.4% in categorizing survival outcomes. This high accuracy underscores the model's proficiency in handling and interpreting complex data patterns associated with ALS. However, CNNs are inherently dependent on large datasets to ensure robustness and avoid overfitting. The necessity for extensive training data highlights a key limitation: without sufficient data, CNNs risk developing models that may not generalize well to new, unseen cases. This challenge necessitates careful data management and augmentation strategies to fully leverage the potential of CNNs in ALS research, ensuring that models remain accurate and reliable across diverse patient populations and data scenarios[4].

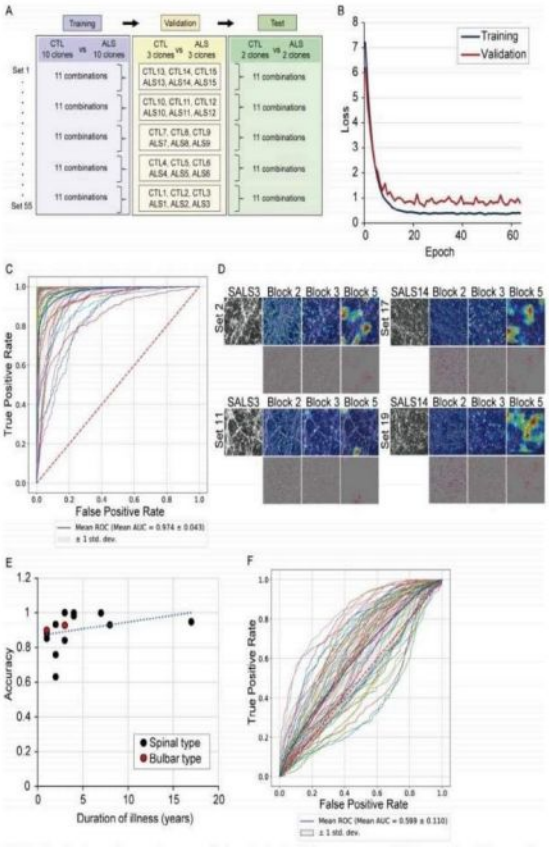


FIGURE 2: Classification performance in amyotrophic lateral sclerosis (ALS) motor neurons. (A) Strategy of model construction of learning, validation, and test. (B) Loss function of learning. There was no increase in loss even in the late stage of learning. (C) Receiver operating characteristic (ROC) curve of classification for healthy controls and ALS subjects from 55 test datasets. The average curve is shown by the thick blue line. (D) Saliency maps for ALS motor neurons by gradient-weighted class activation mapping (Grad-CAM) in upper row and guided Grad-CAM in lower row. The saliency map was centered around neurites in block 2 and soma of motor neurons in block 3. (E) Correlation of the duration of illness and accuracy of the classification. The accuracy of each ALS sample averaged in each set is shown. The positive correlation between the duration of illness and accuracy is presented; n = 15, Spearman coefficient $r = 0.61$, $p = 0.018$. Duration of illness refers to the period from the onset of the disease to the collection of somatic cells for iPSC generation. Correlation of accuracy was not seen between the bulbar type and spinal type of ALS. (F) The random forest classifier resulted in low classification accuracy using these features. Using the same sample set as deep learning experiments, area under the curve (AUC) for ALS diagnosis by the random forest was approximately 0.60. CTL = control.

2.2.5 SUPPORT VECTOR MACHINES WITH RECURSIVE FEATURE ELIMATIONIMINATION (RFE)

Support Vector Machines (SVM) are advanced

classifiers that determine the optimal hyperplane to effectively separate different classes in a high-dimensional space, thereby maximizing the margin between them. By incorporating Recursive Feature Elimination (RFE) into SVM, known as SVM-RFE, the model's performance is further enhanced. SVMRFE systematically evaluates the importance of each feature, removing the least relevant ones and retaining those that contribute most to model accuracy. This process improves both the model's predictive performance and its interpretability.the application of SVM-RFE has yielded outstanding results. For instance, when applied to blood biomarker data, SVMRFE achieved a classification accuracy of 94%. This exceptional performance reflects the method's ability to manage complex and high-dimensional data effectively, pinpointing the most significant biomarkers associated with ALS. By focusing on the most relevant features, SVM-RFE enhances the model's capability to provide accurate predictions and insights into disease progression, thereby supporting better understanding and management of ALS.

3. RESULTS AND DISCUSSION

3.1 CLINICAL DATA-BASED MODELS

Random Forest (RF) models have proven highly effective in predicting ALS progression using clinical data from the PRO-ACT dataset, achieving accuracies of up to 79%. Their strength lies m managing complex, non-linear relationships among clinical variables, such as ALS Functional Rating Scale-Revised (ALSFRS-R) scores and Forced Vital Capacity (FVC) measurements. This performance surpasses Generalized Linear Models (GLMs), which achieved only 66% accuracy, highlighting RFmodels' supenor ability to handle highdimensional and diverse clinical data. By aggregating the predictions of multiple decision trees, RF models enhance robustness and reduce overfitting, offering more accurate and reliable forecasts of ALS progression and survival. This capability underscores the potential of RF models to improve ALS prognosis and patient management by providing valuable insights from clinical data[2].

3.2 MRI BASED MODELS

The integration of MRI data with clinical variables has markedly improved prediction accuracy for ALS progression. Logistic Ridge Regression models, which combine MRI features such as cortical thickness and fractional anisotropy with clinical variables, achieved an accuracy of 79.17% for predicting 18-month survival. This represents a notable enhancement compared to models using MRI data alone, which attained an accuracy of 77.08%. The increased accuracy underscores the added value of incorporating both imaging and clinical data, providing a more comprehensive and nuanced view of ALS progression[4]. By combining these data sources, the models can capture a fuller spectrum of disease characteristics, leading to more precise predictions and better insights into how clinical and neuroimaging features interact to influence disease outcomes[3].

3.3 BLOOD BIOMARKER MODELS

Support Vector Machines with Recursive Feature Elimination (SVM-RFE) have demonstrated exceptional performance in classifying ALS patients compared to those with Lower Motor Neuron Disease (LMND) using blood biomarker data. With a classification accuracy of 94%, SVM-RFE's success is largely due to its effective feature selection process, which prioritizes the most informative biomarkers and minimizes dataset noise[1]. By systematically eliminating less relevant features, SVM-RFE enhances the model's ability to accurately distinguish between ALS and LMND, showcasing the significant Potential of biomarker data in refining predictive models. This high level of performance underscores the value of integrating advanced machine learning techniques with detailed biomarker information to improve diagnostic accuracy and advance our understanding of diseasespecific biomarkers[8].

3.4 DEEP LEARNING MODELS

Deep Leaming models, especially Convolutional Neural Networks (CNNs), have achieved impressive results in predicting survival categories in ALS using MRI data, with an accuracy of 84.4%. This high performance highlights CNNs' capability to uncover complex patterns and relationships within large datasets, thanks to their deep architecture and sophisticated feature extraction processes. However, the effectiveness of CNNs is constrained by their reliance on extensive datasets to avoid overfitting and ensure generalizability. This limitation emphasizes the critical need for larger and more diverse datasets to fully exploit the potential of deep learning models in ALS research[4]. By addressing this need, researchers can enhance the models' ability to provide accurate predictions and insights, ultimately improving understanding and management of ALS[3].

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4. FUTURE DISCUSSION

The integration of clinical, neuroimaging, and biological data in machine learning models offers significant promise for enhancing ALS prognosis. Future research should focus on several key areas :Expanding Datasets Many current studies are constrained by limited sample sizes, particularly for neuroimaging and biomarker data. Collaborative efforts among ALS research centers can facilitate the creation of larger datasets, which will enable the development of more robust and generalizable models. Larger datasets will also allow for more granular analysis, including sub-group variations within the ALS patient population. Model Generalization While existing Models show prom1smg results, they need extensive validation across diverse patient populations and clinical settings. External validation is crucial to assess the generalizability of predictive models and ensure their applicability in real-world scenarios. This includes testing models across different demographics, disease stages, and healthcare environments to enhance their robustness and reliability. Incorporating Longitudinal Data ALS is a progressive disease, and many current models rely heavily on baseline data, which may not fully capture the disease's progression over time[l]. Future models should incorporate longitudinal data to track disease dynamics over extended periods, improving prediction accuracy by accounting for changes in disease status and progression.Hybrid Models Combining different machine learning techniques, such as integrating Random Forest with Support Vector Machines or incorporating deep learning with traditional statistical methods, could lead to hybrid models that leverage the strengths of each approach. These models could effectively integrate data from multiple modalities-including clinical data, neuroimaging, and biomarkers-providing a more comprehensive and accurate picture of ALS progression. Personalized Medicine Approaches Advances in ML and AI can be utilized to develop personalized prediction models based on individual patient profiles. This approach would tailor predictions to specific patient characteristics, improving the accuracy and relevance of forecasts. Incorporating genetic, lifestyle, and environmental factors into predictive models could further refine individual prognosis and treatment strategies[8].

III. CONCLUSION

Based on the review of various algorithms, Support Vector Machines (SVM) with Recursive Feature Elimination (RFE) emerges as the best-performing algorithm in the context of ALS progression prediction[2]. It achieved the highest classification accuracy of 94% when applied to blood biomarker data, thanks to its ability to effectively select and focus on the most relevant features while reducing noise[l]. This high Accuracy, combined with the interpretability of the selected features, makes SVM-RFE particularly valuable in the research setting.However, Random Forest (RF) also proved highly effective in handling clinical and neuroimaging data, with accuracies ranging from 79% to 83%, demonstrating its robustness in integrating diverse data sources. While RF models may not outperform SVM in certain areas, their ability to handle high-dimensional data and their resilience to overfitting make them a strong contender, especially in clinical settings. The integration of these machine learning techniques with comprehensive datasets, along with further refinement in hybrid and personalized approaches, would pave the way for more accurate and reliable ALS progression models[4].

REFERENCES

[l] "Machine learning and brain computer interface approaches in prognosis and individualized care strategies for individuals with amyotrophic lateral sclerosis A systematic review" by Stephanie Yen Nee Kew,Siew-Ying Mok,Choon-Hian Goh.

[2] "RandomForest4Life: A Random Forest for predicting ALS disease progression" By Torsten Hothom & Hans H. Jung.

[3] "Prediction Model of Amyotrophic Lateral Sclerosis by Deep Leaming with Patient Induced Pluripotent Stem Cells"By Keiko Imamura,Yuichiro Yada,Yuishin Izumi.

[4] "Deep learning predictions of survival based on MRI in amyotrophic lateral sclerosis" by Hannelore K. van der Burgha, Ruben Schmidta,Henk-Jan Westenenga.

[5] "Performance of Ridge Logistic Regressionand Decision Tree in the Binary classification" by Marji,Samingun Handoyo.

[6] "Amyotrophic lateral sclerosis: moving towards a new classification system" by Ammar Al-Chalabi, Orla Hardiman, Matthew C Kiernan, Adriano Chio, Benjamin Rix-Brooks, Leonard H van den Berg.

[7] "Identifying and predicting amyotrophic lateral sclerosis clinical subgroups: a population-based machine-learning study" by Faraz Faghri, Fabian Brunn, Arrant Dadu.

[8] "Prognosis for patients with amyotrophic lateral sclerosis development and validation of a personalised prediction model" by Henk-Jan Westeneng, Thomas P A

Debray, Anne E Visser, Ruben PA van Eijk, James PK Rooney, Andrea Calvo, Sarah Martin.

[9] "Amyotrophic Lateral Sclerosis disease progression model,Amyotrophic Lateral Sclerosis Frontotemporal Degenerat" by R.Gomeni,M.Fava.

[1O] "Predictors of diagnostic delay in amyotrophic lateral sclerosis: a cohort study based on administrative and electronic medical records data, Amyotrophic Lateral Sclerosis Frontotemporal Degenerat" by Francesca, S. Ariam1a, L. Giancarlo, P.E. Federica.

[11] "Predicting disease progression in amyotrophic lateral sclerosis, Ann. Clin.Transl. Neurol." By A.A.Taylor, C.Fournier,L.Wang.

[12] "Prognosis for patients with amyotrophic lateral sclerosis: development and validation of a personalised prediction model, Lancet Neurol" by H.JWesteneng ,T.P.Debray,R.P.van Eijk.

[13] "Survival prediction in Amyotrophic lateral sclerosis based on MRI measures and clinical characteristics, BMC Neurol." By C.Schuster,O.Hardiman,P.Bede